

Functional & Performance Testing Template

Model Performance Test

Date	29 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	

Functional Testing

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass/Fail
FT-01	Applicant Input Validation	Enter valid and invalid applicant details	Valid data accepted, invalid data rejected with error message	Works as expected	Pass
FT-02	Numeric Input Validation	Enter valid and invalid values for income, loan amount, loan term	Accepts valid values and displays validation errors for invalid values	Works as expected	Pass
FT-03	Loan Eligibility Prediction	Enter complete applicant information and click Predict	System generates loan approval/rejection result	Prediction displayed successfully	Pass
FT-04	Model Loading Check	Start Flask application and load saved XGBoost model	Model loads successfully without errors	Model loaded successfully	Pass
FT-05	Data Preprocessing Validation	Submit records with missing values and categorical values	Missing values handled and categorical values encoded correctly	Preprocessing completed successfully	Pass

FT-06	Result Display Verification	Submit applicant details and view result page	Approval/Rejection result displayed correctly	Result displayed correctly	Pass
FT-07	Database Storage Check	Submit application and verify database records	Applicant details and prediction results stored successfully	Data stored successfully	Pass

Performance Testing

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass/Fail
PT-01	Response Time Test	Submit application and measure prediction response time	Response generated within 3 seconds	Average response time: 1.8 seconds	Pass
PT-02	Concurrent Request Test	Submit multiple prediction requests simultaneously	System handles multiple requests without failure	Requests processed successfully	Pass
PT-03	Dataset Processing Test	Train model using complete dataset	Training completes successfully without interruption	Training completed successfully	Pass
PT-04	Model Prediction Speed Test	Run prediction for multiple applicants	Prediction generated quickly for all applicants	Predictions generated successfully	Pass
PT-05	Application Load Test	Access application from multiple browsers/users	Application remains stable and responsive	Stable performance observed	Pass
PT-06	Database Performance Test	Store and retrieve prediction records repeatedly	Data operations complete without delay	Database responded normally	Pass